2 Data transmission

1 a Explain the term data packet.

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b Data packets contain a header. Name **three** components that you would expect to find in a header.

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c Data packets also have a payload and a trailer. Explain these two terms.

payload: .....

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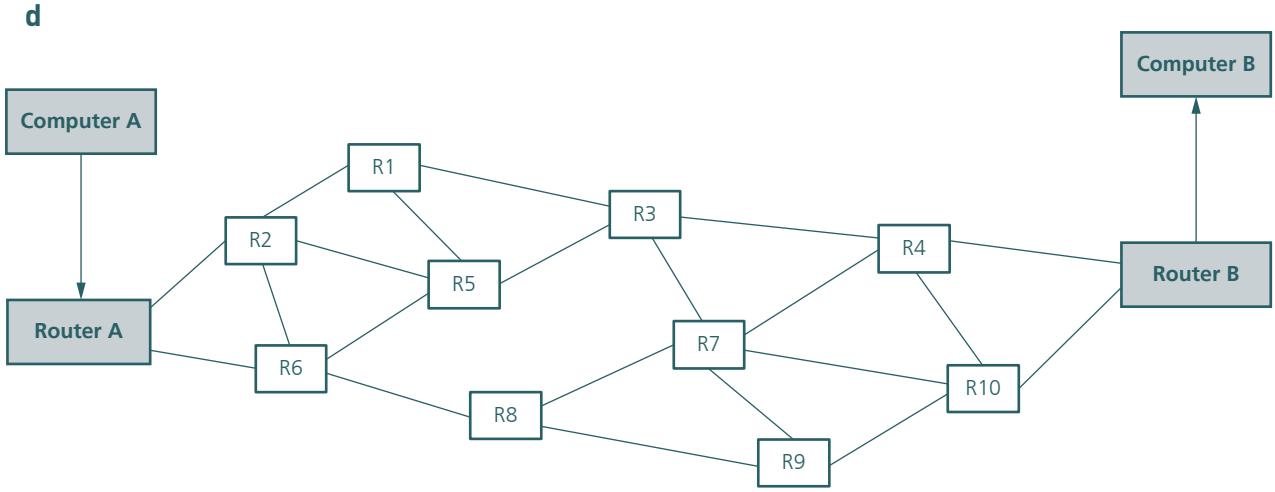
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trailer: .....

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i Using the above diagram, explain how packet switching works when sending a 500 KiB file from computer 'A' to computer 'B'.

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ii Using the diagram, explain why data packets need to be reassembled at computer 'B'.

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2 Packet switching is used to send packets of data over a network.

a Give **three** benefits of using packet switching.

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b Give **three** disadvantages of using packet switching.

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c Data packets can get lost due to 'bouncing' around from router to router and never reaching their destination.

Explain the problems caused by data packets 'bouncing'.

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3 a Data transmission depends on data direction and how many bits of data can be sent at a time. Complete the table below to show what type of data transmission is being described in each case. Tick (✓) two boxes for each description.

Description of transmission taking place	Simplex (✓)	Half-duplex (✓)	Full-duplex (✓)	Serial (✓)	Parallel (✓)
Data sent one bit at a time down a single wire in one direction only					
Data sent 16 bits at a time down 16 wires in both directions, but not at the same time					
Data sent 16 bits at a time down 16 wires in both directions simultaneously					
Data sent 8 bits at a time down 8 wires in one direction only					
Data sent one bit at a time down a single wire in both directions simultaneously					
Data sent one bit at a time down a single wire in both directions, but not at the same time					

b Five statements about serial and parallel data transmission are made in the table below. By placing a tick (✓) in the appropriate column, select which statements refer to serial transmission and which statements refer to parallel transmission.

Statements	Serial (✓)	Parallel (✓)
Transmission method used by the memory bus inside a computer		
Data can be skewed (out of synch) when travelling over long distances		
Least expensive of the two types due to fewer hardware requirements		
Most appropriate if data is time-sensitive; for example, when live streaming where faster transmission rate is essential		
Suffers from less risk of external interference		

4 a Explain what happens when a device is plugged into a computer using one of the USB ports.

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b Seven statements about USB connections are made in the table below. By ticking (✓) the appropriate box, indicate which statements are true and which statements are FALSE.

Statements	True (✓)	False (✓)
USB uses a protocol that allows for error-free data transmission between device and computer		
USB can support any cable length between device and computer		
USB uses serial data transfer		
USB connections can transfer data using half-duplex or full-duplex		
It is difficult to add more USB ports to a computer using USB hubs		
USB data transfer rates are much faster than, for example, Ethernet connections		
There is no need for devices to have a power source because USB cable supplies 5V of power		

5 a A system uses even parity. Indicate whether the following bytes would fail or pass an even parity check:

1	1	0	0	0	0	1	1

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1	0	1	1	0	1	1	0

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0	0	0	1	1	1	1	1

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b Explain why parity checks are used.

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c Nine bytes of data were transmitted from one computer to another computer. Even parity was used by both systems. An additional byte, called the parity byte was also sent at the end of the transmission.

The following table shows the nine bytes and parity byte following transmission.

	Parity bit	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
Byte 1	1	1	1	0	1	1	1	0
Byte 2	1	0	0	0	0	1	0	0
Byte 3	0	1	1	1	0	0	1	0
Byte 4	0	1	1	1	1	0	1	1
Byte 5	1	1	0	0	0	1	1	0
Byte 6	0	1	1	0	1	1	0	1
Byte 7	1	0	0	1	0	0	0	0
Byte 8	0	1	1	1	1	1	0	1
Byte 9	0	0	0	0	1	0	0	1
Parity byte:	0	0	1	1	1	1	0	0

i One of the bits has been transmitted incorrectly. Indicate which bit is incorrect by giving its bit number and byte number:

bit number: .....

byte number: .....

ii Explain how you arrived at your answer to part **c i**.

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iii Write down the corrected byte.

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iv Describe a situation where a parity check wouldn't identify which bit had been transmitted incorrectly.

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v Name and briefly describe another method to check if data has been transmitted correctly.

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6 a Explain how the checksum system is used to identify any errors during data transmission.

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b Explain how echo checks are used to check data following data transmission.

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c Explain how automatic repeat requests (ARQs) work during data transmission.

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7 Check digits are used to check data during data entry.

a Give **four** types of error that check digits can identify.

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b ISBN-13 is a type of check digit standard.

i The following algorithm generates a check digit for a 12-digit ISBN book code:

1 add all odd-numbered digits together

2 add all even-numbered digits together and then multiply the result by 3

3 add together the results for steps 1 and 2, and then divide the result by 10

4 take away the remainder of the division from 10, leaving the check digit

Showing your working, calculate the check digit for the following 12-digit ISBN code:

9 781 471 86867

ii The following algorithm checks that the check digit in a 13-digit code is correct:

1 add all odd-numbered digits together, including the check digit

2 add all even-numbered digits together and then multiply the result by 3

3 add together the results for steps 1 and 2, and then divide the result by 10

4 the check digit is correct if the remainder is zero

Showing your working, use the above algorithm to check whether or not the check digit in the following ISBN-13 code is correct:

9 781 471 80721 8

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8 For each of the following ten questions, choose which of the five options corresponds to the correct response.

a What is meant by the term ciphertext when used in encryption? Circle the correct answer.

A an encryption or decryption algorithm

B a message before it has been encrypted

C a type of session key

D another name for plaintext

E text following an encryption algorithm

b Which one of the following statements about asymmetric encryption is true? Circle the correct answer.

A uses a public key only

B uses both a public key and a private key

C always uses 64-bit encryption

D the value of the private key is symmetrical

E uses private key only

c In encryption, which of the following is the term used to describe the message before it is encrypted? Circle the correct answer.

A simpletext

B plaintext

C notext

D ciphertext

E firsttext

d Which of the following is the biggest disadvantage of using symmetric encryption? Circle the correct answer.

A it is very complex and time consuming

B it is rarely used anymore

C the value of the key reads the same in both directions

D it only works on computers with older operating systems

E there is a security problem when transmitting the encryption key

e Which of the following is the correct name for a form of encryption in which both the sender and the recipient use the same key to encrypt and decrypt? Circle the correct answer.

A symmetric key encryption

B asymmetric key encryption

C public key encryption

D same key encryption

E block cipher encryption

f Which of the following is the final number in a code, calculated from all the numbers in the code, whose purpose is to find errors in data entry? Circle the correct answer.

A parity check

B checksum

C cyclic redundancy check

D parity bit

E check digit

g Which of the following is a form of error detection that makes use of a system of acknowledgements and timeouts? Circle the correct answer.

A automatic repeat request

B echo check

C check digit

D parity bit

E cyclic redundancy check

h Which of the following methods uses an extra bit added to a byte to ensure it contains an even number of 1s or odd number of 1s? Circle the correct answer.

A cyclic redundancy check

B parity check

C checksum

D check digit

E echo check

i Which of the following uses a calculated value which is sent after a block of data: the receiving computer also calculates the value from the block of data and compares the values? Circle the correct answer.

A parity check

B check digit

C packet switching

D checksum

E automatic repeat request

j Which of the following describes the check where the receiving computer sends back a copy of the data to the sending computer to allow it to compare the data? Circle the correct answer.

A echo check

B automatic repeat request

C checksum

D parity check

E check digit

9 Some students sat a computer science exam. A number of their responses were incorrect.

Explain why the following eight responses given by the students are incorrect:

a 'A data packet header contains the MAC address of the sending computer and the receiving computer.'

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b 'Half-duplex refers to the fact that four bits of data (half a byte) can be sent from "A" to "B" along four single wires.'

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c 'Because USB connections can be fitted in any way, they have become the industry standard.'

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d 'Odd parity means a binary number has an odd value, for example, 01110001 (i.e. 113).'

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e 'Even parity means a binary number has an even number of zeros.'

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f 'Check digits are used to make sure data has been transmitted without errors.'

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g 'A checksum is calculated by adding up all the 1-bits in a byte, and sending this value at the end of the byte.'

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h 'Asymmetric encryption makes use of the same key to encrypt and decrypt an encoded message.'

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10 Five statements are shown on the left and eight computer terms are shown on the right in the following diagram.

Draw lines to connect each statement to the correct computer term.

Error checking method where data is sent to another device and is then sent back to the sender; if the two sets of data are different then there has been an error during transmission	Symmetric encryption
	Encryption
Data is made unreadable to anyone without the necessary keys to unlock the encoded data	Cipher text
	Echo check
A form of encryption which makes use of a public key and a private key to encrypt and decrypt coded messages	Full-duplex
Transmission error check that makes use of positive acknowledgement and timeout following data transmission	Asymmetric encryption
	Half-duplex
Type of data transmission where data can be sent in both directions at the same time	Automatic repeat request

11 Use the following word/phrase list to complete the paragraph below. Each word or phrase may be used once, more than once or not at all.

» algorithm	» decrypt	» matching pairs	» private
» asymmetric encryption	» document	» matching private	» public
» cipher text	» encrypted	» mathematically	» public key
» confidential	» matching document	» plain text	» symmetric encryption

..... makes use of a ..... key and a ..... key. Suppose Asif and Karl both work for the same company, and Asif wants to send a confidential ..... to Karl.

Asif and Karl's computers both use the same ..... to generate their own ..... of keys; these keys are ..... linked, but cannot be derived from each other. Karl now sends his ..... to Asif. Asif now uses Karl's ..... to encrypt the document he wishes to send to Karl.

Asif then sends his ..... document back to Karl. Karl uses his ..... key to unlock Asif's ..... and ..... it.